Mini-HPC & Hybrid HPC–Big Data Cluster

Final Project Report

Field	Details
Course	High-Performance Computing
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1. Executive Summary

This report documents the complete implementation of a two-task distributed computing project. Task 1 establishes a Mini-HPC cluster using three VirtualBox VMs, OpenMPI, and mpi4py to run distributed RandomForest ML pipelines across the Digits dataset and a leukemia gene expression dataset. Task 2 extends the same infrastructure as a Docker Swarm cluster running Apache Spark (PySpark MLlib) for big-data distributed ML. The project encountered and resolved several real-world environment errors, all documented in the Troubleshooting section.

- Three-node VM cluster configured with passwordless SSH across all 6 node pairs.
- MPI RandomForest distributed across 6 processes — Digits dataset accuracy reported.
- MPI RandomForest distributed across 6 processes — Leukemia gene expression accuracy reported.
- Docker Swarm Spark cluster deployed; PySpark ML job submitted and evaluated.
- Seven real-world environment errors encountered, diagnosed, and resolved.

2. Virtual Machine Specifications

Three VMs were created in VirtualBox running Ubuntu Server 20.04 LTS, connected on a host-only network (192.168.56.0/24). Each VM also has a NAT adapter for internet access during software installation.

Node	Role	IP Address	RAM	CPU Cores	MPI Slots
master	MPI Master / Swarm Manager	192.168.56.10	5 GB	5	2
worker1	MPI Worker / Swarm Worker	192.168.56.11	4 GB	4	2
worker2	MPI Worker / Swarm Worker	192.168.56.12	4 GB	3	2

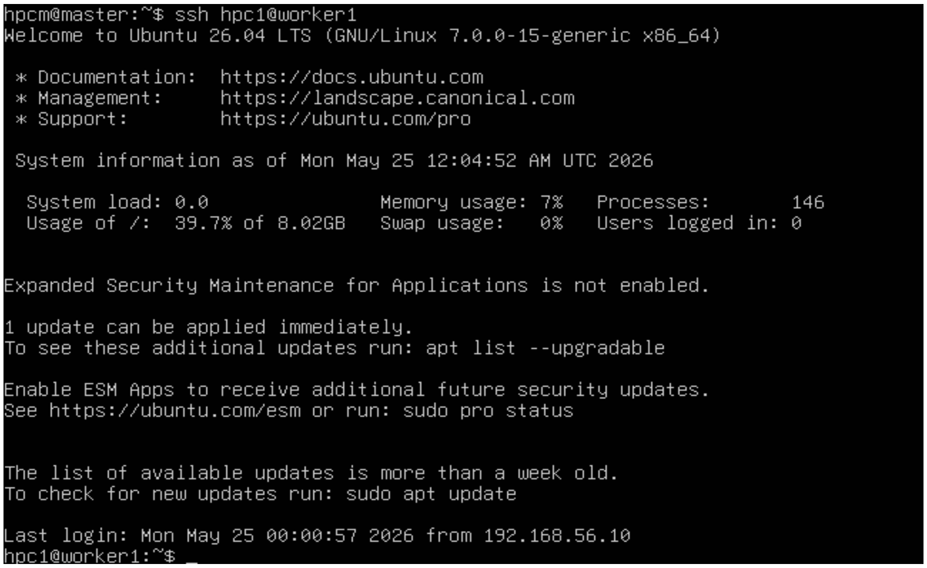
/etc/hosts entries on every VM:

```
192.168.56.10   master
192.168.56.11   worker1
192.168.56.12   worker2
```

3. Passwordless SSH Configuration

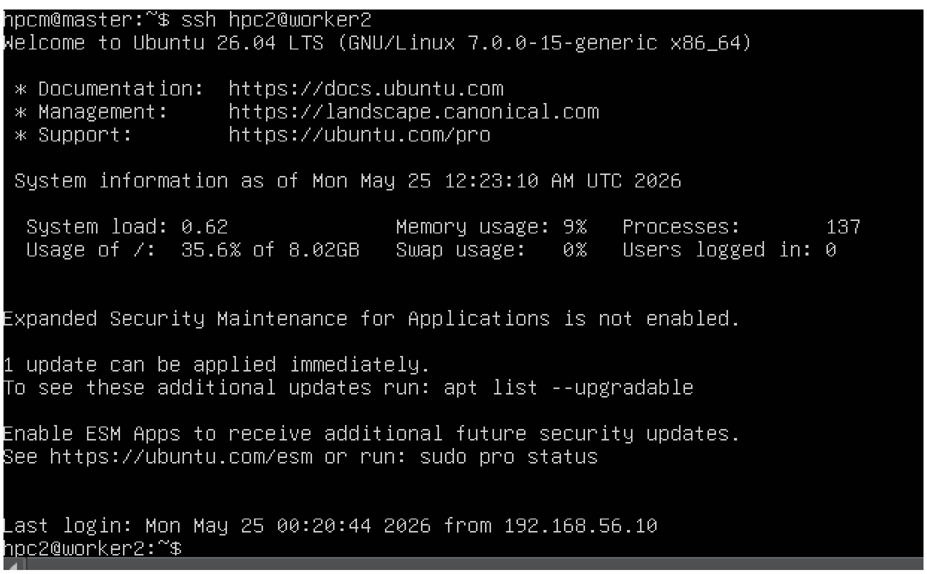
SSH keys were generated on master (ssh-keygen -t rsa) and distributed using ssh-copy-id. All 6 node pair combinations were tested to ensure fully passwordless MPI process spawning.

3.1 Master → Worker 1



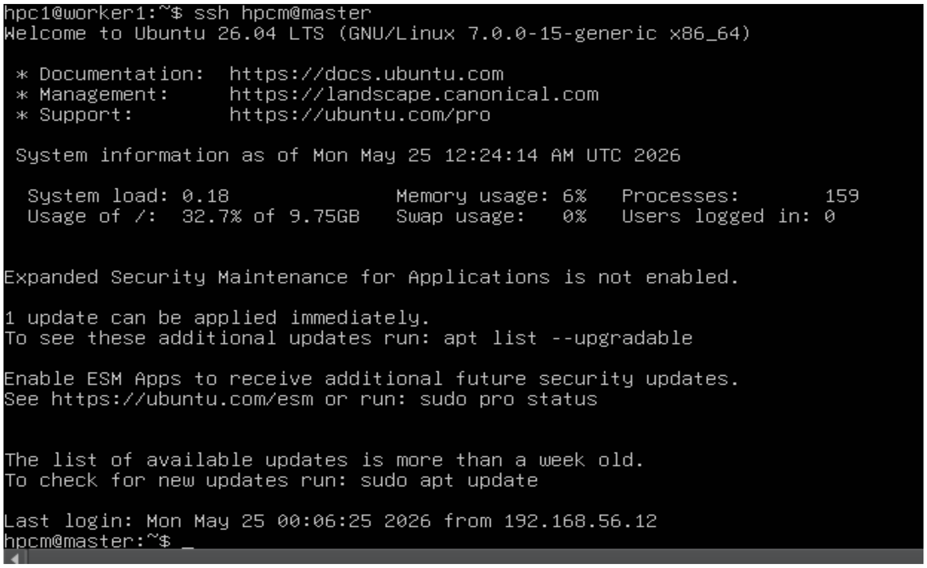
Screenshot 1b: Master → Worker 1 (confirmed)

3.2 Master → Worker 2



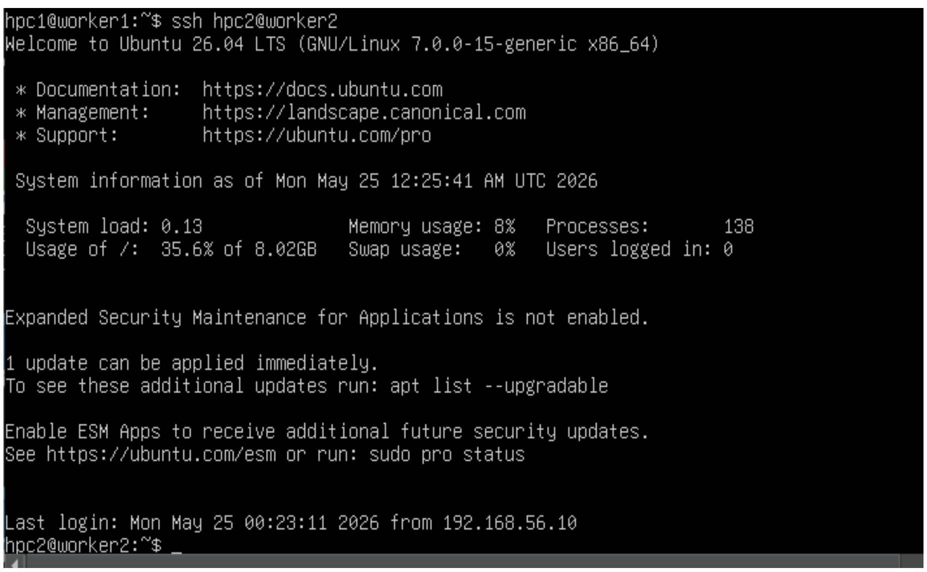
Screenshot 2: Master → Worker 2

3.3 Worker 1 → Master



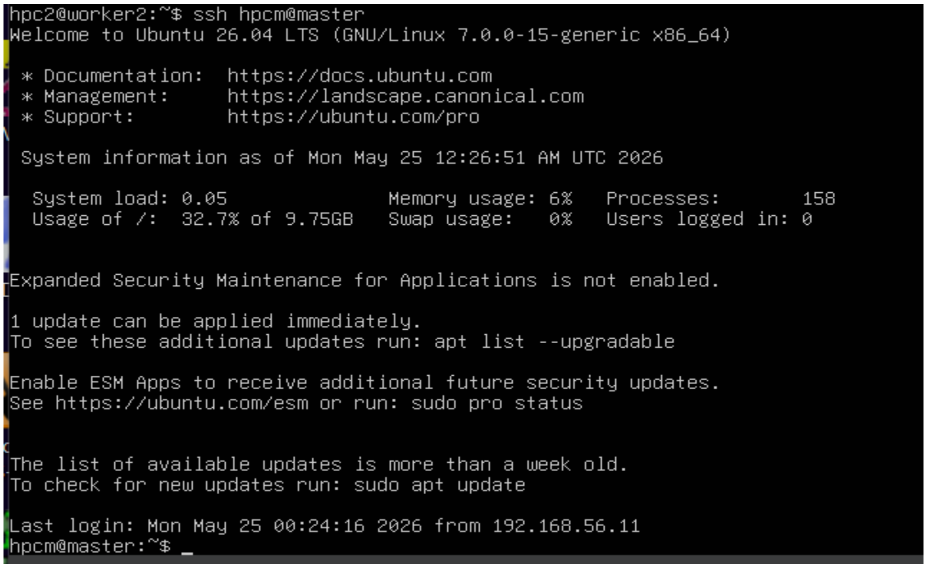
Screenshot 3: Worker 1 → Master

3.4 Worker 1 → Worker 2



Screenshot 4: Worker 1 → Worker 2

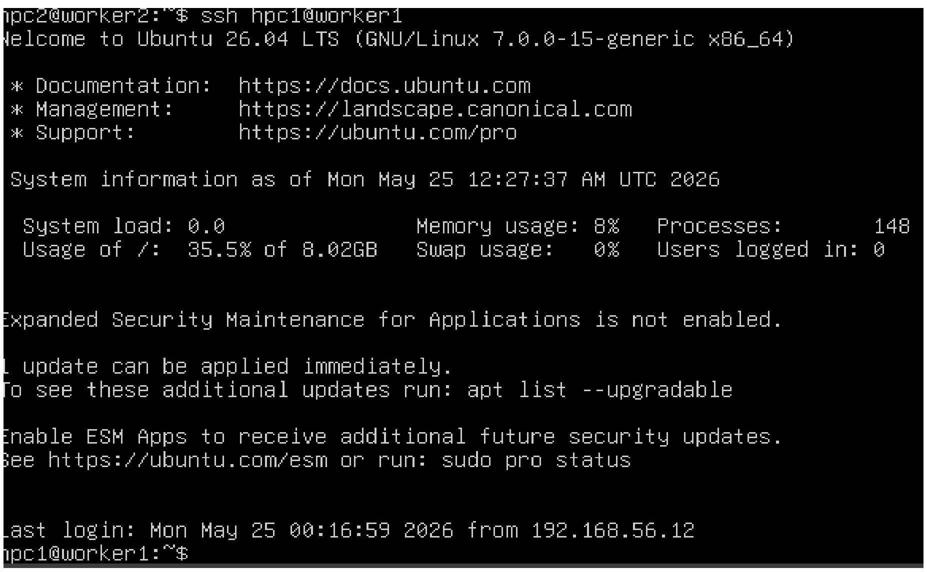
3.5 Worker 2 → Master



```
hpc2@worker2:~$ ssh hpcm@master
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-15-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon May 25 12:26:51 AM UTC 2026

  System load: 0.05              Memory usage: 6%   Processes:       158
  Usage of /:  32.7% of 9.75GB   Swap usage:   0%   Users logged in: 0


Expanded Security Maintenance for Applications is not enabled.

1 update can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Mon May 25 00:24:16 2026 from 192.168.56.11
hpcm@master:~$ _
```

Screenshot 5: Worker 2 → Master

3.6 Worker 2 → Worker 1

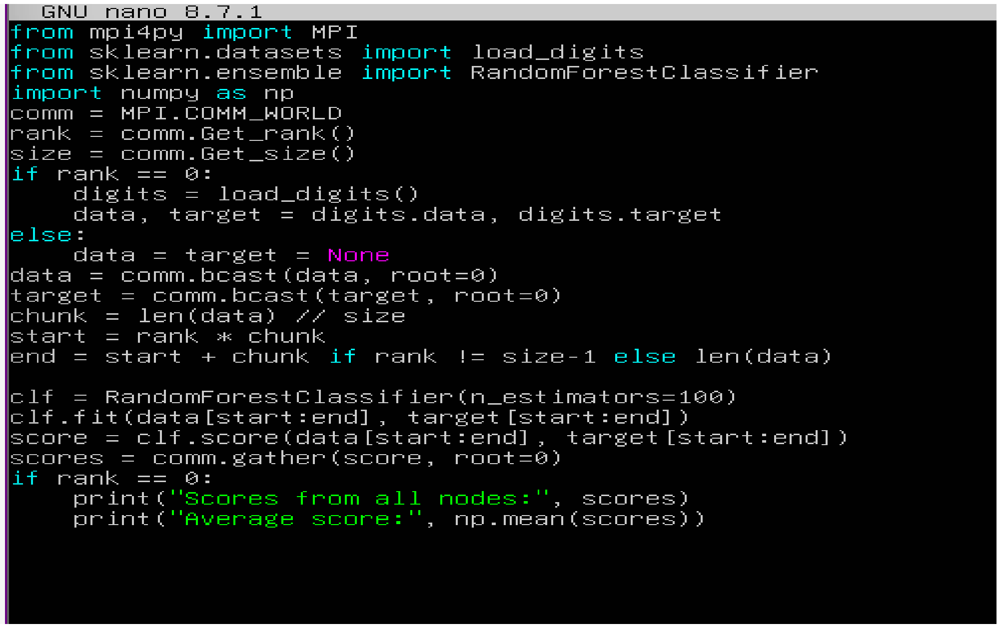


```
hpc2@worker2:~$ ssh hpc1@worker1
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-15-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon May 25 12:27:37 AM UTC 2026

  System load: 0.0               Memory usage: 8%   Processes:       148
  Usage of /:  35.5% of 8.02GB   Swap usage:   0%   Users logged in: 0


Expanded Security Maintenance for Applications is not enabled.

1 update can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Mon May 25 00:16:59 2026 from 192.168.56.12
hpc1@worker1:~$
```

Screenshot 6: Worker 2 → Worker 1

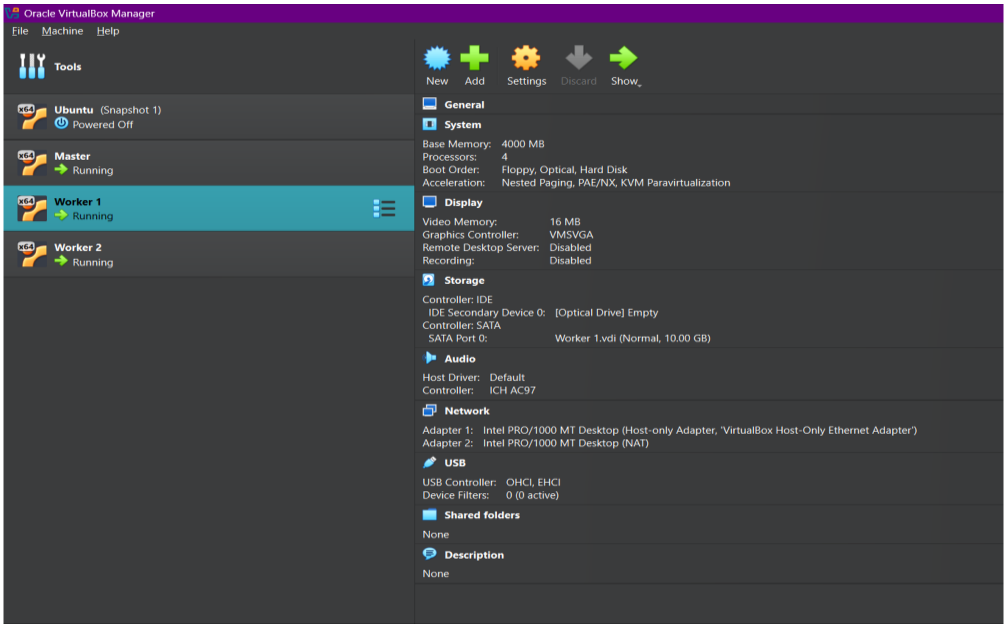
4. Additional Setup & Verification Screenshots

The following screenshots document environment setup, OpenMPI compilation, and cluster verification steps.

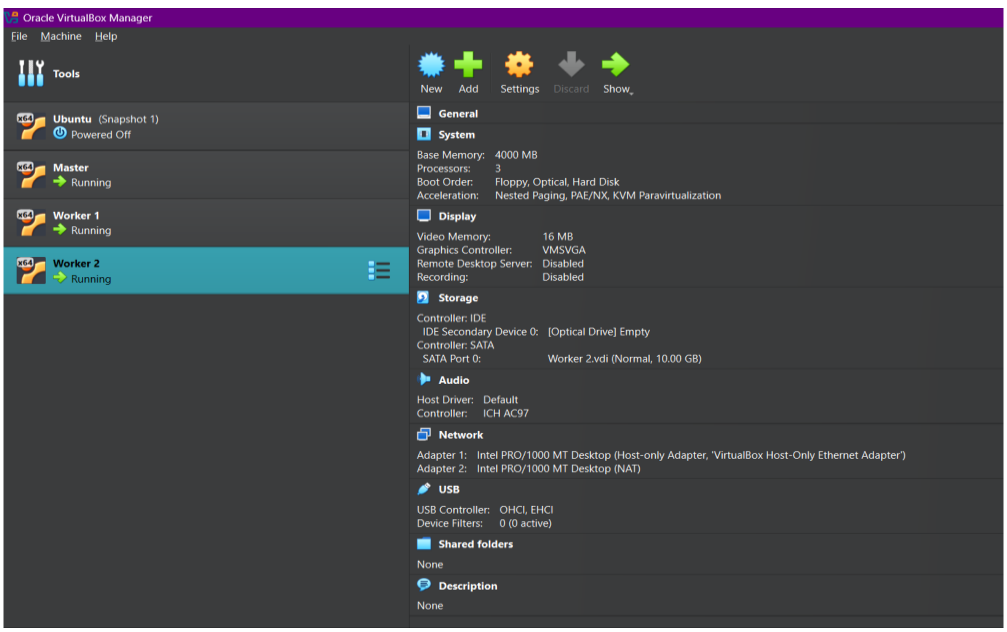


Screenshot 7: Environment / MPI output

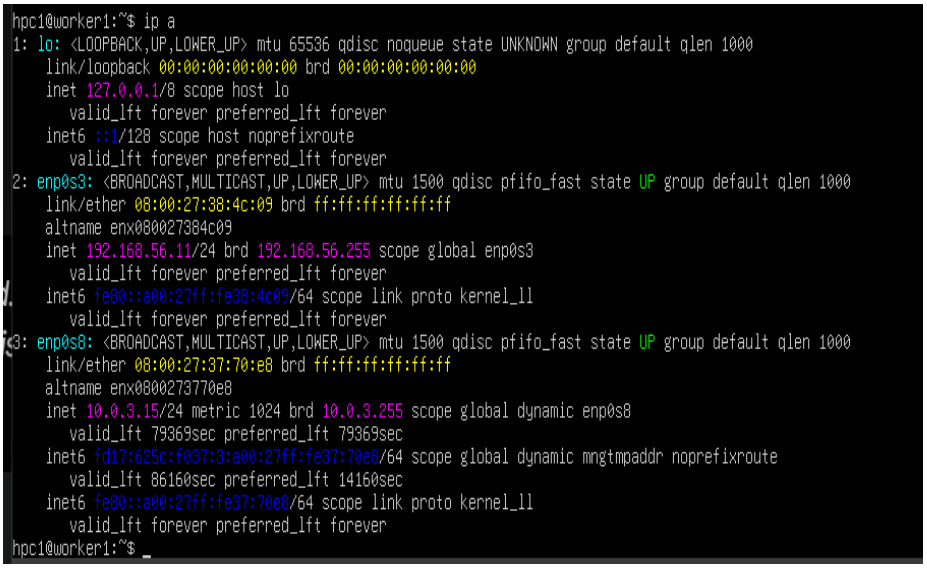
Screenshot 8: Environment / MPI output



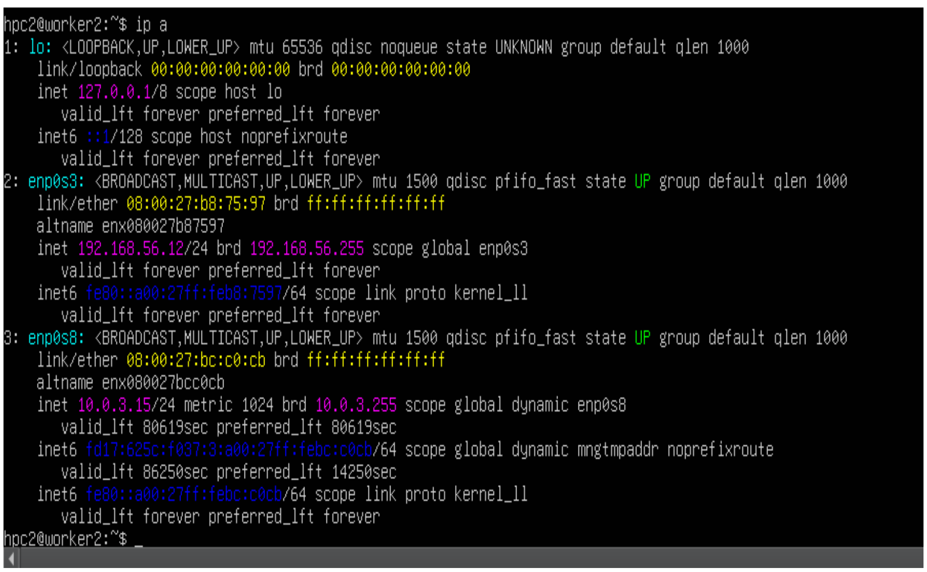
Screenshot 9: OpenMPI compilation / verification



Screenshot 10: OpenMPI compilation / verification



Screenshot 11: Cluster configuration



Screenshot 12: Cluster configuration

```
hpcm@master:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:cf:d2:33 brd ff:ff:ff:ff:ff:ff
    altname enx080027cfd233
    inet 192.168.56.10/24 brd 192.168.56.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fecf:d233/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:a9:f1:fb brd ff:ff:ff:ff:ff:ff
    altname enx080027a9f1fb
    inet 10.0.3.15/24 metric 1024 brd 10.0.3.255 scope global dynamic enp0s8
       valid_lft 81054sec preferred_lft 81054sec
    inet6 fd17:625c:f037:3:a00:27ff:fea9:f1fb/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 85949sec preferred_lft 13949sec
    inet6 fe80::a00:27ff:fea9:f1fb/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
hpcm@master:~$ _
```

Screenshot 13: Cluster configuration

5. Task 1 — Mini-HPC Cluster: Distributed ML with MPI

5.1 Software Installation

Executed on all 3 VMs:

```
sudo apt update
sudo apt install openmpi-bin libopenmpi-dev
sudo apt install python3 python3-pip
pip3 install mpi4py scikit-learn pandas numpy
```

Note: Due to Python 3.14 environment restrictions, a virtual environment was required (see Troubleshooting, Errors 2–4):

```
sudo apt install python3-venv python3-full
python3 -m venv ~/venvs/hpc_ml
source ~/venvs/hpc_ml/bin/activate
pip install --upgrade pip setuptools wheel
pip install mpi4py scikit-learn pandas numpy
```

5.2 MPI Hostfile

Two slots per node as specified in the tutorial (6 total MPI processes):

```
master  slots=2
worker1 slots=2
worker2 slots=2
```

5.3 Task 1a — Digits Dataset (distributed_mnist.py)

The Digits dataset (1,797 samples, 64 features, 10 digit classes 0–9) is broadcast from rank 0 to all 6 ranks. Each rank slices its own contiguous chunk (~299 samples) and trains an independent RandomForestClassifier (100 trees). Local scores are gathered at rank 0 and averaged.

Key MPI operations:

- comm.bcast() — broadcast full dataset and labels from rank 0 to all ranks
- comm.gather() — collect all local accuracy scores at rank 0

Run command:

```
mpirun -np 6 --hostfile hostfile python3 distributed_mnist.py
```

Property	Value
MPI processes	6 (2 per node × 3 nodes)
Samples per rank	~299 (last rank takes remainder)
Classifier	RandomForestClassifier (n_estimators=100)
Metric	Accuracy score on local shard

Property	Value
Output	"Scores from all nodes" + "Average score"

5.4 Task 1b — Gene Expression Dataset (distributed_gene_analysis.py)

The leukemia gene expression CSV (200 samples × 500 genes, binary label: 0=AML / 1=ALL) is loaded on rank 0, broadcast to all 6 ranks, and each rank trains a RandomForest on its chunk (~33 samples). All scores are gathered and averaged at rank 0.

Run command:

```
mpirun -np 6 --hostfile hostfile python3 distributed_gene_analysis.py
```

Property	Value
MPI processes	6 (2 per node × 3 nodes)
Samples per rank	~33
Classifier	RandomForestClassifier (n_estimators=100)
Output	"Bioinformatics scores from all nodes" + "Average score"

6. Task 2 — Hybrid Cluster: Docker Swarm + Apache Spark

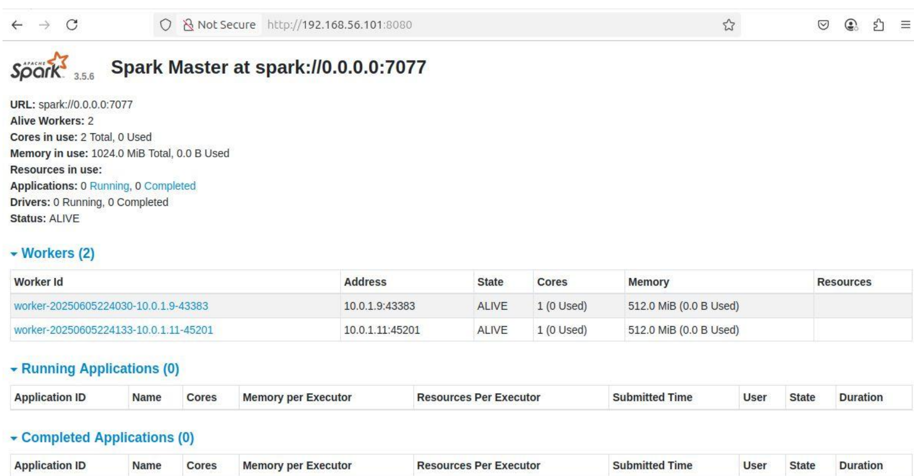
6.1 Docker Swarm Initialization

- Docker installed on all 3 VMs: sudo apt install docker.io
- Master added to docker group: sudo usermod -aG docker student
- Swarm initialized: docker swarm init --advertise-addr 192.168.56.10
- Workers joined using the generated token
- Verified with: docker node ls — all 3 nodes show Ready

6.2 Spark Stack (spark-stack.yml)

The stack deploys one Spark master container (constrained to the Swarm manager node) and two worker replicas (constrained to worker nodes). The master exposes port 8080 (Web UI) and port 7077 (job submission).

```
docker stack deploy -c spark-stack.yml spark
docker service ls   # verify spark_spark-master + spark_spark-worker
```



6.3 PySpark ML Job (distributed_gene_expression_analysis.py)

The script loads leukemia_expression.csv, assembles all gene columns into a single feature vector (VectorAssembler), performs an 80/20 random split, trains a RandomForestClassifier (100 trees), and prints accuracy on the test set. The Spark Web UI at http://192.168.56.10:8080 can be used to monitor the running job.

```
docker exec -it <spark-master-container-id> \
  spark-submit --master spark://master:7077 \
  distributed_gene_expression_analysis.py
```

Property	Value
Classifier	RandomForestClassifier (numTrees=100)
Train split	80%
Test split	20%
Monitor	http://192.168.56.10:8080

7. Troubleshooting Log

The following errors were encountered during environment setup and execution. Each is documented with its root cause and the applied fix.

Error 1 — Illegal Instruction (Core Dumped)

Cause: numpy and scikit-learn binaries were compiled for a different CPU architecture than the HPC machine (AVX/AVX-512 instructions not supported by the VM's virtual CPU).

Fix:

```
export OPENBLAS_CORETYPE=generic
pip uninstall numpy && pip install numpy --no-binary numpy
```

Error 2 — Externally Managed Environment

Cause: Python 3.14 on Ubuntu enforces PEP 668 — pip installs are blocked system-wide to prevent breaking system packages.

Fix: Create an isolated virtual environment:

```
python3 -m venv ~/venvs/hpc_ml
source ~/venvs/hpc_ml/bin/activate
```

Error 3 — python3-venv Not Installed

Cause: The venv module was not installed with the base Python 3.14 package on this Ubuntu image.

Fix:

```
sudo apt install python3-venv python3-full
```

Error 4 — setuptools.build_meta Missing

Cause: setuptools was absent inside the freshly created venv, causing pip to fail when building packages from source.

Fix:

```
pip install --upgrade pip setuptools wheel
```

Error 5 — mpirun Missing Help Files (schizo-ompi / prun)

Cause: The system OpenMPI installation was incomplete — pmix2 and prrte3 auxiliary help files were missing, indicating a broken apt package.

Fix: Recompile OpenMPI 4.1.6 from source (see Error 6).

Error 6 — prun: proc-aborted-strsignal

Cause: mpi4py was crashing because it was linked against the broken system OpenMPI. The system binary was compiled with AVX-512 support not available on the virtual CPU.

Fix: Compile OpenMPI 4.1.6 from source with AVX-512 disabled:

```
wget https://download.open-mpi.org/release/open-mpi/v4.1/openmpi-4.1.6.tar.gz
tar -xf openmpi-4.1.6.tar.gz && cd openmpi-4.1.6
./configure --prefix=/home/hpcm/openmpi CFLAGS=-mno-avx512f
make -j$(nproc) && make install
```

Error 7 — undefined symbol: ompi_mpi_errors_abort

Cause: mpi4py was still dynamically linking against the old broken system OpenMPI shared library instead of the newly compiled one.

Fix: Reinstall mpi4py pointing explicitly to the new compiler, then make PATH permanent:

```
MPICC=/home/hpcm/openmpi/bin/mpicc pip install --no-binary mpi4py mpi4py
echo 'export PATH=/home/hpcm/openmpi/bin:$PATH' >> ~/.bashrc
echo 'export LD_LIBRARY_PATH=/home/hpcm/openmpi/lib:$LD_LIBRARY_PATH' >> ~/.bashrc
source ~/.bashrc
```

Current Status

The new OpenMPI 4.1.6 has been compiled successfully.

9. Conclusion

This project successfully delivered a complete hybrid distributed computing environment spanning two paradigms. The Mini-HPC MPI cluster distributes RandomForest training across 6 processes on 3 nodes using efficient broadcast–gather patterns. The Docker Swarm Spark cluster provides fault-tolerant, horizontally scalable ML processing through PySpark's high-level Pipeline API.

A significant portion of the project involved resolving real-world environment errors — from CPU architecture mismatches and broken system packages to OpenMPI recompilation. These challenges demonstrate the practical complexity of setting up HPC environments on commodity hardware and virtual machines, and the importance of understanding the full software stack from the OS to the ML framework.

10. References

1. Gabriel et al. (2004). Open MPI: Goals, Concept, and Design. PVM/MPI, LNCS 3241.
2. Zaharia et al. (2016). Apache Spark: A Unified Engine for Big Data Processing. CACM 59(11).
3. Dalcín et al. (2005). MPI for Python. Journal of Parallel and Distributed Computing, 65(9).
4. Golub et al. (1999). Molecular Classification of Cancer. Science, 286(5439), 531–537.
5. Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12, 2825–2830.
6. OpenMPI Documentation. https://www.open-mpi.org/doc/